Memo

Authors: Wan Chi Kao, Chengyuan Liu

Date:2026/10/13 Subject: Product recommendations and customer groups

Data Quality

We combined 1,067,371 transaction records and removed duplicate, invalid and non-product entries. We also adjusted purchases for matched returns. The final training data covers 5,463 customers and 3,684 products. We kept the last two months separate for testing, so future purchases were not used to build the recommendations.

Recommendation Results

We tested all methods on the same 1,906 customers. A recommendation counted as successful if a recommendation counted as successful if the customer later bought at least one of its ten products for the first time. Our main SVD model succeeded for 23.50% of customers, compared with 17.37% for best sellers, an improvement of 6.14 percentage points. PCA reached 22.67%. In the purchase-value comparison, binary reached 24.50%, while raw quantity reached 14.27%. We chose the “log” version as our main method before looking at any test results, so the comparison is fair. The simple “bought or not” version did slightly better and is the first thing we would try next. Using raw quantities did worse than best sellers, because a few wholesalers who buy thousands of units dominate the numbers.

Risks and Next Steps

Return matching is approximate because the data does not directly link every return to its original purchase. The evaluation covers existing customers and products with enough purchase history, so it does not measure performance for new customers or rare products. It also measures future purchases, not whether recommendations caused extra sales. The next step is a small controlled trial to check whether the recommendations improve sales before wider use.

Method

The full customer-by-product table would take 161 MB. We keep 100 summary numbers per customer and per product instead, which takes about 7 MB. Building the model took about 0.2 seconds, and suggestions for all 5,463 customers took about 1 second, so the job fits easily in a nightly run. What we lose is the summary does not keep exact quantities, such as how many units each customer bought. It keeps the buying patterns, which products tend to be bought by the same customers, and that is what the suggestions need.

Storage and Speed

With 100 components, SVD took 0.2035 seconds to train, compared to 0.3140 seconds for PCA. The stored customer and product factors only needed about 6.98 MiB, and generating recommendations for all 5,463 customers took about 1.09 seconds. These results show that the model can reduce storage and generate recommendations quickly in our experiment.

Customer Segments

We found five customer groups from their buying patterns: Baking & Home, Mixed Shoppers, Home Decoration, Bags & Lunch, and Baking & Tea. Marketing can plan one campaign for each. About 93% of customers are in the UK, so country does not separate the groups much; Bags & Lunch has the most customers from abroad.